Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Technical Report - English (Svarah) and Telugu (IndicTTS Telugu)

GitHub repository: https://github.com/karankumar211/IIITH-ASSIGNMENT

1. Implementation Approach

Two separate FastSpeech 2 models were trained, one for English and one for Telugu, starting from the public reference implementation by ming024 (github.com/ming024/FastSpeech2). Each language has its own code folder, configuration (config.yaml), checkpoints and results. Alignment, preprocessing, training and inference were run independently.



Figure 1. Processing pipeline used by the English and Telugu FastSpeech 2 systems.

1.1 Components reused from the reference implementation

- Model: 4-layer FFT encoder and 6-layer FFT decoder (hidden size 256, 2 attention heads, convolution kernels [9, 1], filter size 1024), variance adaptor with duration, pitch and energy predictors and a length regulator, mel-spectrogram decoder with postnet (about 35.3 million parameters).
- Pitch and energy are modelled at phoneme/grapheme level, normalised and quantised into 256 linear bins; the variance adaptor is trained with ground-truth durations from forced alignment.
- Loss: mel (before/after postnet) + pitch + energy + duration terms. Optimiser: Adam with 4,000-step warm-up and step decay.
- Vocoder: pre-trained universal HiFi-GAN supplied with the reference code; audio features use 22,050 Hz, FFT 1024, hop 256, 80 mel bands and fmax 8 kHz.
- Forced alignment: Montreal Forced Aligner (MFA) provides unit durations required by FastSpeech 2.

1.2 Modifications common to both languages

- Compatibility fixes for current libraries: librosa keyword-only arguments, removal of deprecated NumPy aliases and float32 casting of pitch, energy and mel tensors to avoid PyTorch type mismatch errors.
- MFA-3 TextGrid handling: empty intervals are treated as silence (sp).
- Language-specific preprocess.py scripts write split configuration files, run MFA alignment, extract features and recreate train/validation splits from the selected utterance lists, preventing overlap.
- Language-specific inference.py scripts synthesize test sentences and generate training/validation loss plots, plus duration, pitch and energy plots and CSV values for each test sentence.
- A duration-scale option (d_control) was added to control speaking rate during inference.

1.3 English-specific modifications

- Alignment uses the pretrained MFA english_us_arpa acoustic model and dictionary.
- ARPAbet phoneme symbols are retained. At inference, g2p_en performs context-aware grapheme-to-phoneme conversion so homographs such as conduct and out-of-vocabulary words can be handled.
- The English model is multi-speaker, using the two speaker labels supplied by Svarah.

1.4 Telugu-specific modifications

- A new symbol set uses Telugu Unicode grapheme tokens (U+0C00-U+0C7F), together with digits, punctuation and silence symbols. Graphemes are used as acoustic units because a suitable Telugu phoneme dictionary and pretrained aligner were unavailable.
- A grapheme dictionary was built from training transcripts, and an MFA acoustic model was trained on the Telugu data using a monophone-only configuration with 20 iterations.
- Text normalisation applies Unicode NFC, removes zero-width joiners and non-Telugu characters; utterances containing digits were excluded from training.
- The Telugu dataset has two speakers; speaker ID 0 is used at inference.

2. Datasets, Subsampling and Preprocessing

2.1 Selection strategy

- English: AI4Bharat Svarah (Hugging Face), containing Indian-accented read and spontaneous speech. Telugu: SPRINGLab IndicTTS Telugu (Hugging Face), derived from the IIT Madras IndicTTS database.
- Utterances were shuffled using fixed seed 42 and selected until approximately five hours of training audio and one hour of validation audio were reached.
- Filters: audio duration 1-11 seconds; English text cleaning (lower-casing, bracketed tags and non-letters removed); Telugu-script filtering; utterances containing digits were excluded because they could not be aligned reliably.
- Each utterance belongs to either training or validation, never both. Exact utterance lists are stored in data_split/<Language>_split.json.

2.2 Resulting data

Language	Dataset	Train utts	Val utts	Train h (selected)	Val h (selected)	Train h after alignment	Val h after alignment	Speakers
English	Svarah	3,380	691	4.71	0.97	3.62	0.75	2 (female, male)
Telugu	IndicTTS Telugu	2,613	525	5.00	1.00	3.51	0.69	2

The selected durations represent raw audio chosen by sampling. The aligned durations are lower because MFA trims leading/trailing silence and removes a small number of utterances that failed alignment (English: 3,370 train / 686 validation; Telugu: 2,601 train / 520 validation).

2.3 Preprocessing steps

- Audio was resampled to 22,050 Hz, peak-normalised and stored as 16-bit WAV; one .lab transcript was created per utterance.
- MFA generated TextGrid alignments and per-unit durations.
- Mel-spectrogram, pitch (PyWorld) and energy features were extracted at phoneme/grapheme level; global pitch/energy normalisation statistics were saved.

3. Training Configuration

Setting	English	Telugu
Total steps	40,000	40,000
Batch size / gradient accumulation	16 / 1	16 / 1
Optimiser	Adam (0.9, 0.98), eps 1e-9	Same
Learning-rate schedule	4,000-step warm-up; ×0.3 decay at 70% and 85%	Same
Checkpoint / validation frequency	Every 10,000 / 1,000 steps	Same
Alignment / units	Pretrained MFA, ARPAbet phonemes	MFA trained on data, graphemes
Speakers	2	2
Hardware	Kaggle NVIDIA T4 GPU	Kaggle NVIDIA T4 GPU

A 300-step smoke test was run first to validate the pipeline; its checkpoint was discarded. Final results use the step-40,000 checkpoint for each language.

4. Loss Curve Analysis

Model	Train total loss (~100 → 40,000)	Validation total loss (~100 → 40,000)	Lowest validation loss
English	9.69 → 0.96	8.52 → 4.12	3.91 (step 1,000)
Telugu	9.92 → 0.82	9.01 → 4.05	3.82 (step 2,000)

- Convergence: training loss falls quickly during the first thousands of steps and then decreases more slowly. Final training mel loss is approximately 0.46 for English and 0.39 for Telugu.
- Overfitting: validation total loss stops improving early. After about 5,000 steps it remains around 4.05-4.13 for both languages, while training loss continues to decrease. No instability or divergence was observed.
- Checkpoint choice: because validation loss is nearly flat after 5,000 steps, later checkpoints are not clearly better on unseen data. The final step-40,000 checkpoint was used.
- The available summaries do not provide a complete component-wise train/validation series for mel, pitch, energy and duration. Therefore, the total-loss gap cannot be attributed quantitatively to one specific variance-adaptor term.

Loss-curve figures: the English and Telugu training_loss.png and validation_loss.png plots are included under Results/<Language>/. The curves show recorded log entries, not every optimizer update.

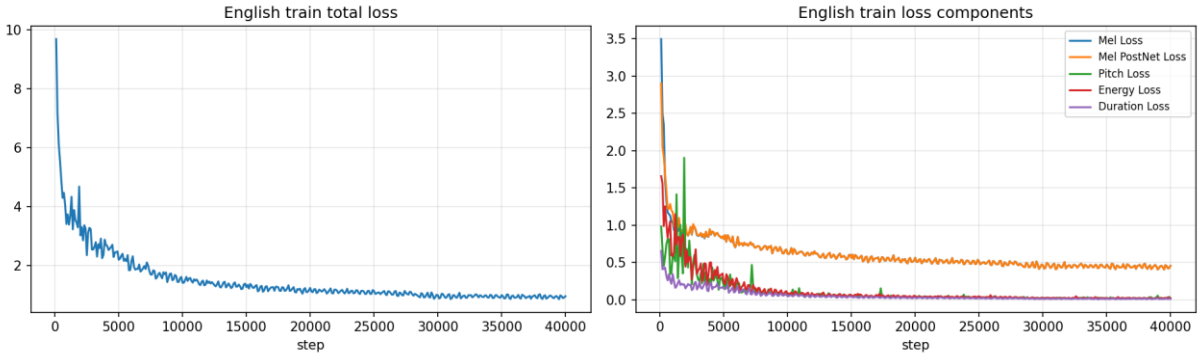


Figure 2. English training loss and loss components.

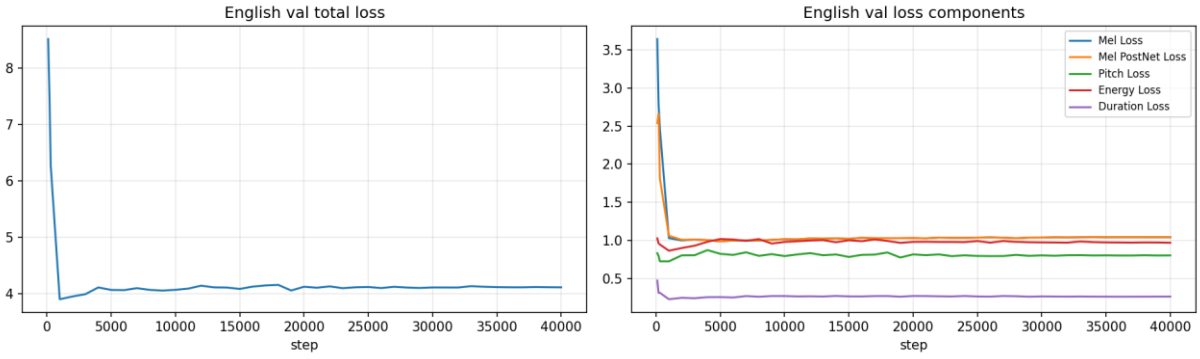


Figure 3. English validation loss and loss components.

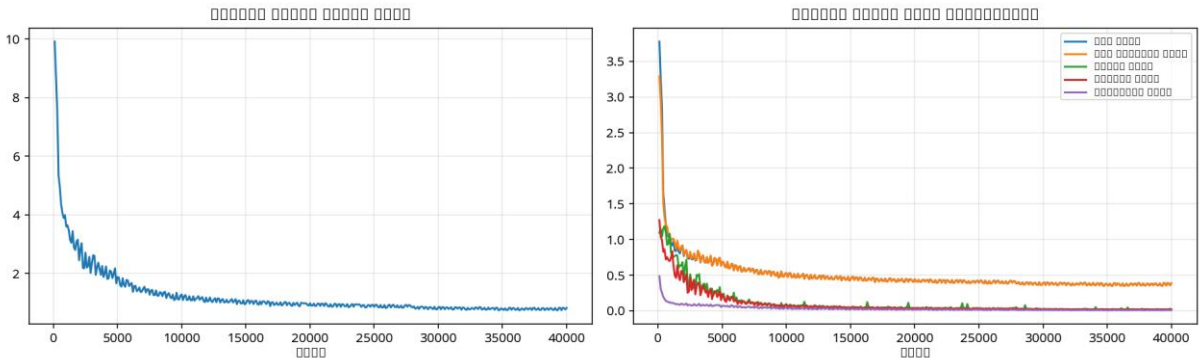


Figure 4. Telugu training loss and loss components.

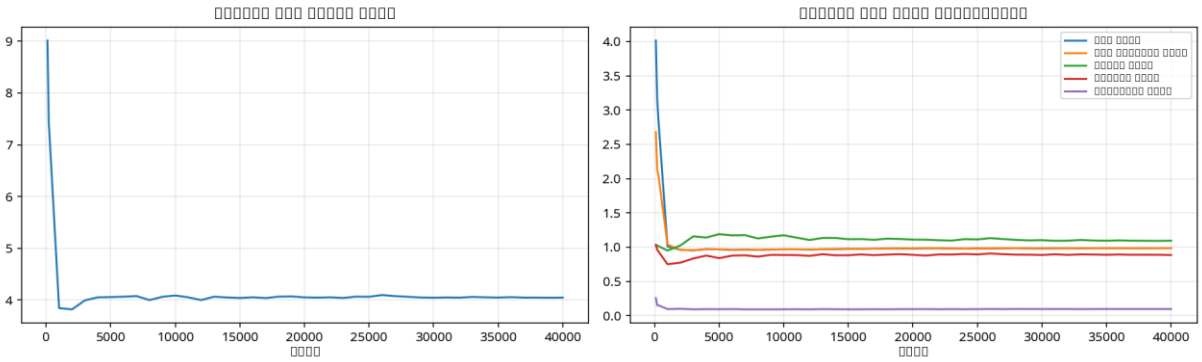


Figure 5. Telugu validation loss and loss components.

5. Generated Speech - Observations per Test Sentence

The final project outputs contain five assignment test audios per language: English E1, E2a, E2b, E3 and E4; Telugu T1, T2a, T2b, T3 and T4. The following observations are based on the generated files and available listening notes; no unsupported word-level pronunciation claims are made.

Duration scaling: the default duration scale produced speech that was noticeably too fast, particularly for longer inputs. The duration control was increased to 1.5 (d_control = 1.5) for final generation. This inference adjustment does not change the trained checkpoint.

ID	Sentence / test focus	Expected difficulty	Observation
E1	Long sentence containing antidisestablishmentarianism	Long rare-word pronunciation and duration allocation	Long input; duration scaling was used to reduce overly fast speech. English output was generally understandable.
E2a	“conduct” used as a noun	Context-sensitive homograph pronunciation	Part of the English evaluation set; no separate sentence-level defect note was retained.
E2b	“conduct” used as a verb	Context-sensitive homograph pronunciation	Part of the English evaluation set; English samples were generally understandable with occasional pronunciation/clarity variation.
E3	Sentence containing the split form “rhyth m”	Robustness to unusual tokenisation	No reliable sentence-specific listening note was retained; exact pronunciation remains unverified.
E4	Long sentence containing date/time and “andaroo”	Number/time handling and unfamiliar word	Long input; duration scaling was used. No separate word-level error note was retained.
T1	Long Telugu sentence	Long-form synthesis and duration allocation	Telugu samples were reported as clear and understandable overall.
T2a	“పట్టు” in the sense of grip	Context-dependent word pronunciation	Telugu samples were reported as clear and understandable overall.
T2b	“పట్టు” in the sense of silk	Context-dependent word pronunciation	Telugu samples were reported as clear and understandable overall.
T3	Telugu sentence containing a split word	Robustness to unusual spacing	No separate sentence-level defect note was retained.
T4	Telugu sentence containing date/time and numbers	Number and time expression handling	No separate sentence-level defect note was retained.

6. Variance Adaptor Analysis (Duration, Pitch, Energy)

For each test sentence, the inference run extracted predicted duration, pitch and energy sequences from the variance adaptor and saved plots and CSV values. Duration predictions determine how many acoustic frames each input unit contributes through the length regulator. Pitch and energy predictions represent learned prosodic contours; their local rises and falls indicate relative variation in model output.

The CSV values are model-output values. Pitch values are available as pitch_hz in the exported analysis tables and energy is available as the exported energy measure. These values describe the inference outputs and should not be interpreted as independently measured ground-truth prosody.

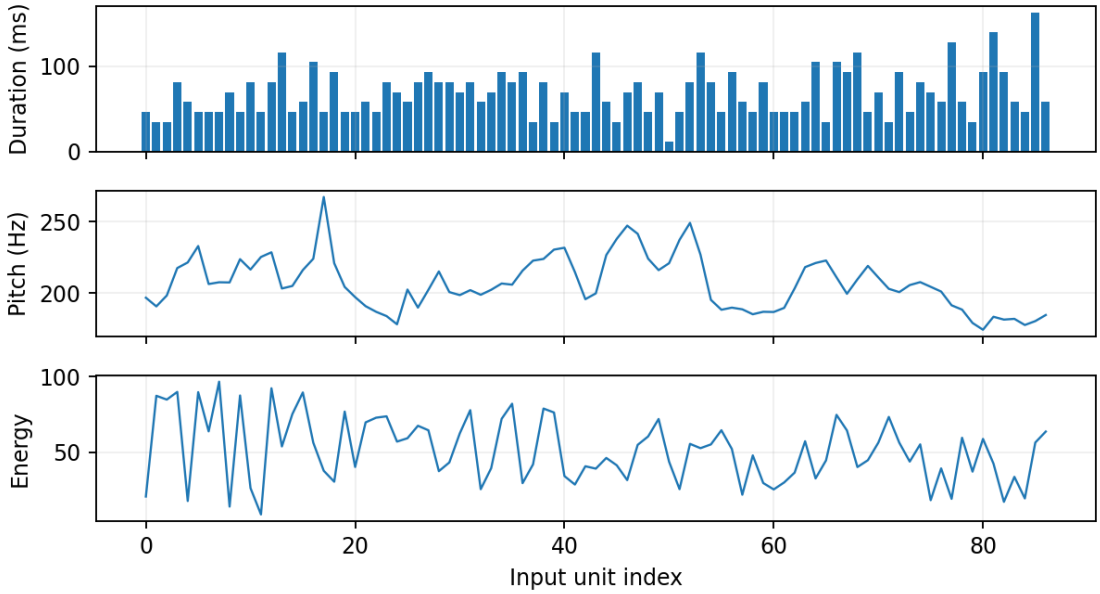


Figure 6. Duration, pitch (F0) and energy predictions for English test E1.

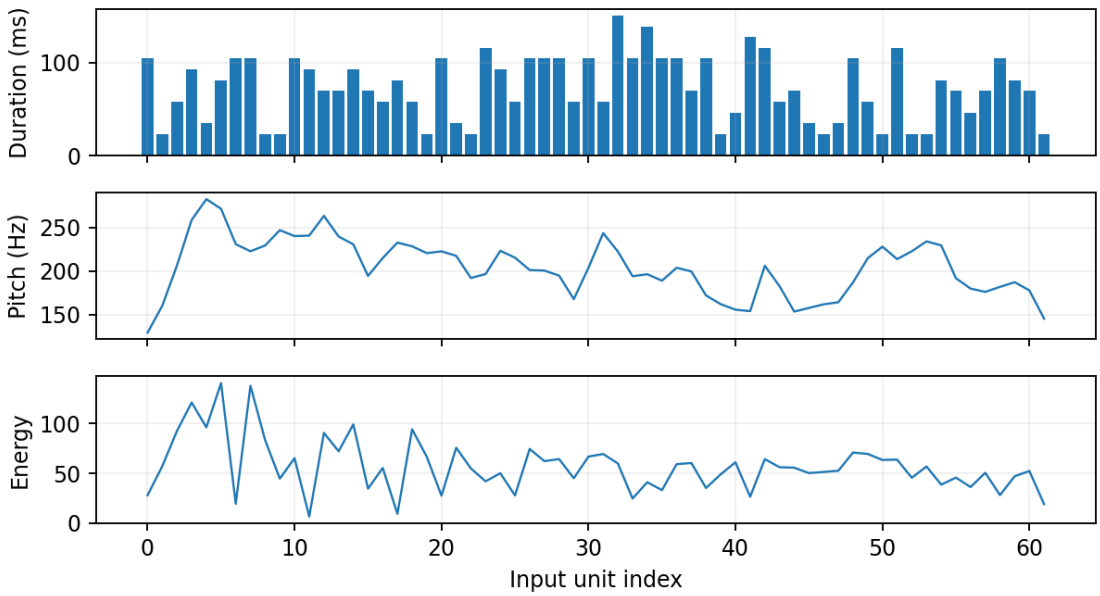


Figure 7. Duration, pitch (F0) and energy predictions for Telugu test T1.

ID	Units	Total duration	Mean unit duration	Pitch range (Hz)	Energy range

E1	87	5.94 s	68.3 ms	174.3-267.5	9.0-96.5
E2a	51	3.95 s	77.4 ms	177.0-232.4	5.2-107.7
E2b	56	4.12 s	73.6 ms	173.3-246.8	5.8-98.5
E3	37	2.75 s	74.4 ms	178.7-228.8	9.7-139.9
E4	79	6.33 s	80.1 ms	176.2-284.5	6.9-138.3
T1	62	4.54 s	73.2 ms	129.3-282.4	6.7-140.8
T2a	28	2.40 s	85.8 ms	171.9-244.5	1.3-136.6
T2b	34	2.33 s	68.6 ms	148.6-241.4	4.2-115.0
T3	29	1.79 s	61.6 ms	154.8-256.5	13.3-109.6
T4	78	6.03 s	77.2 ms	135.0-259.7	9.3-128.7

The plots for all ten test sentences are stored under Results/<Language>/variance_adaptor_plots/{duration,pitch,energy}/. The CSV files provide token-level values. Total duration and mean unit duration are calculated from the exported duration frames and the 22,050 Hz / 256-hop feature setup.

7. Comparison, Limitations and Possible Improvements

- Data quantity: only about 3.5-3.6 hours of effective aligned speech per language remained after trimming; validation loss plateaued early, indicating a substantial generalisation gap.
- English Svarah contains Indian-accented read and spontaneous speech recorded in varied conditions. Telugu IndicTTS is more controlled; dataset and recording differences affect the comparison.
- Alignment: English used a pretrained aligner. Telugu used a monophone grapheme aligner trained on a relatively small dataset, so Telugu unit durations may be less accurate.
- Telugu graphemes are an imperfect substitute for phonemes, especially for conjuncts and inherent vowels, which can limit pronunciation accuracy.
- Training used utterances of 1-11 seconds. Longer test sentences and unseen words, digits or time expressions are challenging and may require stronger text normalisation.
- Possible improvements include using more training data, data augmentation, early stopping or regularisation, a Telugu G2P/phoneme lexicon, a better-trained Telugu aligner, number/time normalisation and a vocoder fine-tuned on target speakers.

8. Reproducibility

- Software: Python 3, PyTorch, librosa, PyWorld, g2p_en, tgt, PyYAML, datasets, matplotlib and Montreal Forced Aligner 3.4.x; see requirements.txt.
- Hardware: Kaggle notebook with an NVIDIA T4 GPU.
- Run preprocessing inside each language folder: python preprocess.py
- Train using the generated configuration: python train.py -p config/<lang>/preprocess.yaml -m config/<lang>/model.yaml -t config/<lang>/train.yaml
- Generate inference outputs: D_CONTROL=1.5 python inference.py --out ../Results/<lang>
- HiFi-GAN: extract hifigan/generator_universal.pth.tar.zip inside hifigan/. Final step-40,000 checkpoints are stored in <lang>/checkpoints/ and can be distributed through Git LFS. Dataset split lists are stored in data_split/<Language>_split.json.

9. Conclusion

Two independent FastSpeech 2 models for English and Telugu were trained to 40,000 steps and used to synthesize the prescribed five test cases per language. Training loss decreased substantially, while validation loss plateaued early, indicating a generalisation gap with the limited aligned speech. Duration scaling improved the speaking rate for longer inputs. The project includes language-specific preprocessing, alignment, training and inference code, separate

checkpoints, loss plots, generated test audios and variance-adaptor analyses. The numerical analysis and listening observations document the outputs without claiming unverified word-level performance.